



Presenter: William Sung

Agenda

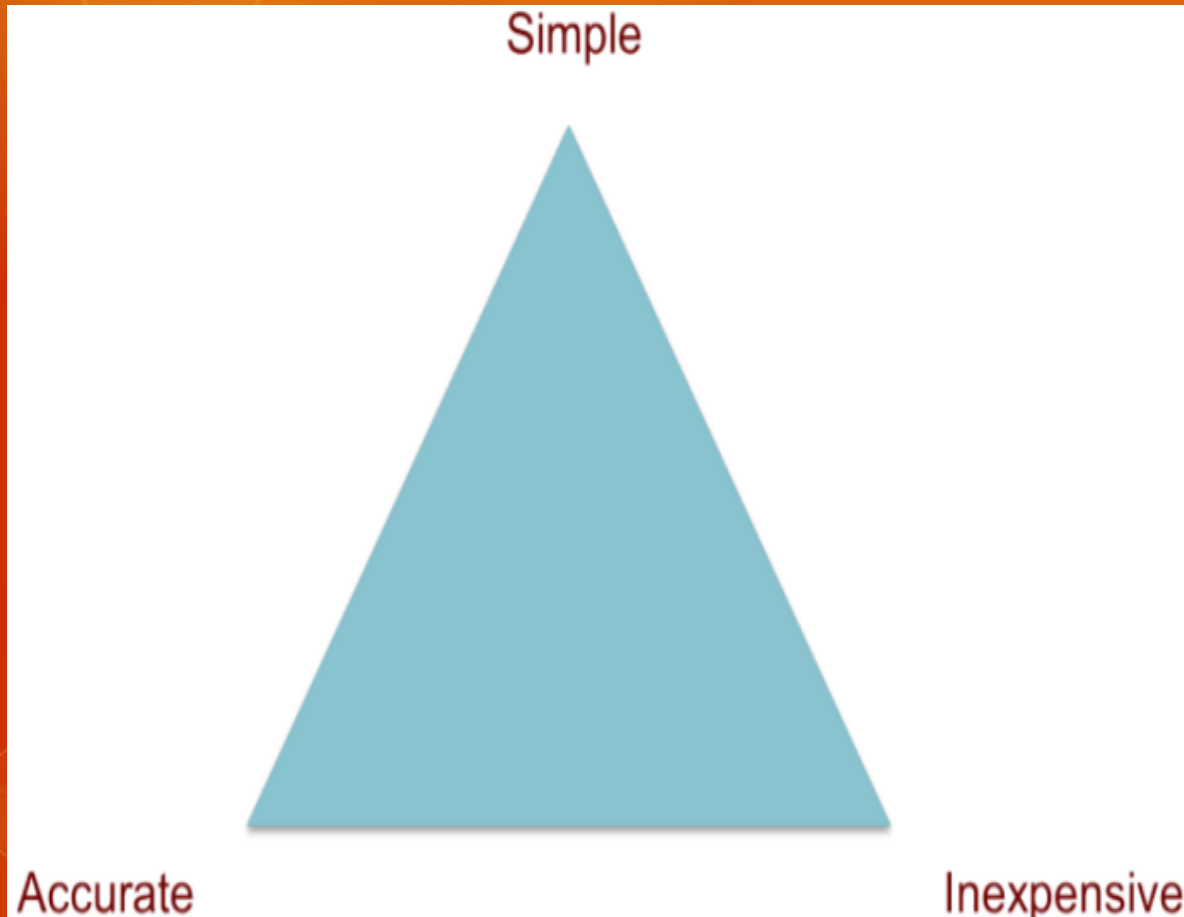
- Introduction of Customer Lifetime Value (CLV)
- Historical CLV
- Predictive CLV
- Conclusion
- Q&A

Introduction

Few things to keep in mind before we start...

- Customer Lifetime Value, known as CLV or LTV
 - In this presentation we will use CLV
- There is no single ultimate way to calculate your CLV
 - Make few tries with different variations
- It is only a tool to test marketing spending and effectiveness
 - Don't over use it, according to online critics, many people mistakenly treat this tool as competitive advantage to create strategies
 - My suggestion: use it as one of your many tools but not the only tool to design your marketing strategy
- Predicting future performance while future is unknown
 - Fluctuating marginal errors, generally 15%(Cohort method) ~ 25%(ARPA method)
 - Usually used to predict next 3-7 years

Intro - Acquiring CLV



Dilemma:

Simple + Accurate
but Costly... (external service)

or

Simple + Inexpensive
but Inaccurate... (DIY)

or

Accurate + Inexpensive
but Complicate... (not easy)

Intro - Two Major Classes of CLV

- **Historical CLV:**

- ARPU / ARPA: Average revenue per account
- Cohort: Advanced version of ARPA, evaluate by grouping customers to segments

- **Predictive CLV:**

- Probabilistic Models
- Pareto/NBD (Negative Binomial Distribution) Models
- Out of current scope due to complexity, we use online templates

Historical CLV

CLV - ARPA

- ARPA: The average revenue per account
- Step 1
 - Alice: $(150 + 50 + 100) / 6 \text{ months} = \50
 - Bob: $(45 + 75 + 100) / 2 \text{ months} = \110
- Step 2
 - $(50 + 110) / 2 \text{ customers} = \80 per customer
- Step 3
 - 12 month value: $\$80 \times 12 = \960
 - 24 month value: $\$80 \times 24 = \1920

*For marketing purpose, remember to at least add profit margin and acquisition cost into account for better accuracy.

Customer Name	Purchase Date	Amount
Alice	January 1, 2013	\$150
Alice	May 15, 2013	\$50
Alice	June 15, 2013	\$100
Bob	May 1, 2013	\$45
Bob	June 15, 2013	\$75
Bob	June 30, 2013	\$100

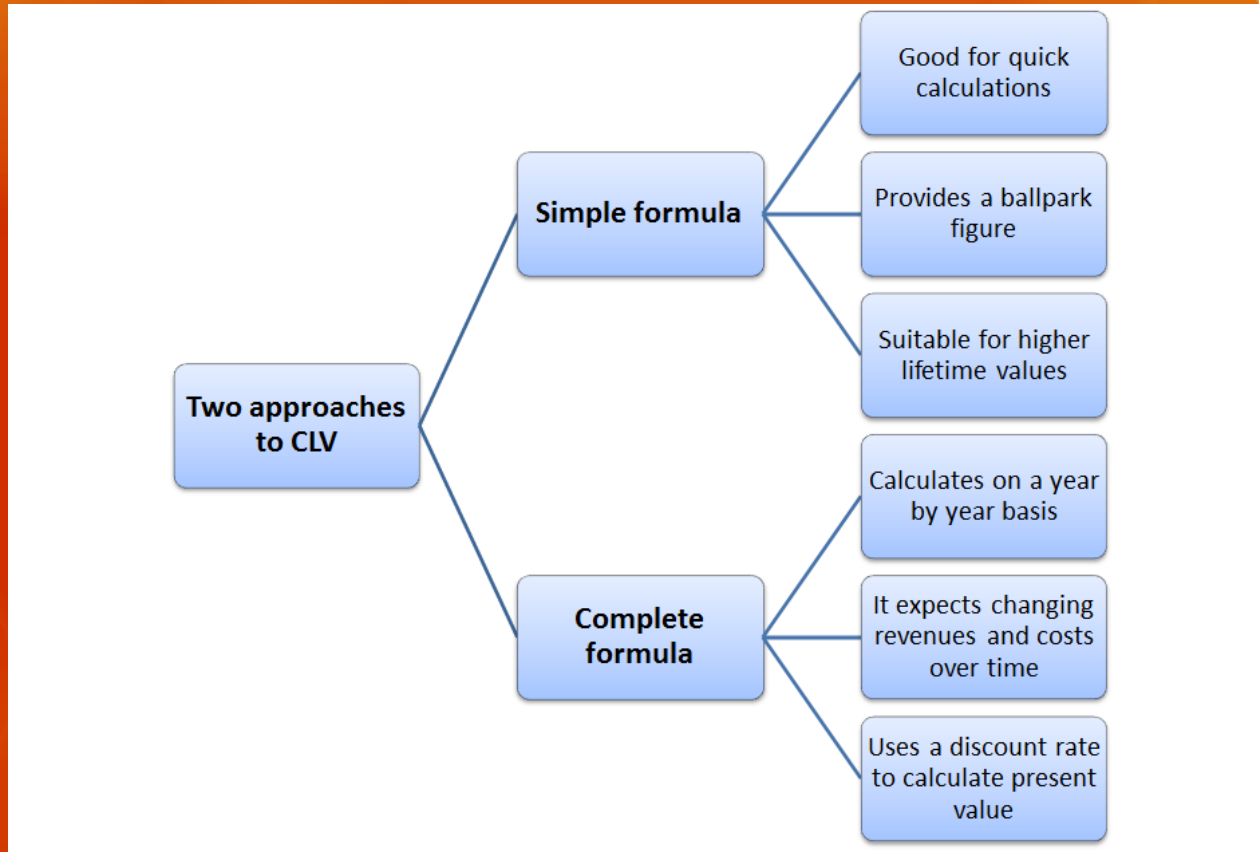
Source: <https://university.custora.com/for-marketers/clv/intermediate/arpv>

CLV - Cohort

- Cohort: a group of customers who share an attribute or a set of attributes.
 - e.g.// new customer, acquired through specific campaign or channel, VIPs
- Limitations: (factors making CLV inaccurate)
 - Not taking customer behavior change into account
 - Different customers have different behaviours, e.g. new vs old customers
 - Outcome could then be misleading
 - Use Cohort method to narrow down possible variations
- When a company is growing fast, expanding product offerings
 - Internally change customer purchasing behaviours
- When market dynamics are changing
 - External factors change customer behaviours
- Not enough data
 - When a business is too young or data duration is too short

Predictive CLV

Two Ways for Predictive CLV



Don't forget you can add cohort into account.

CLV – Variables - I

FOUR major variables to acquire before the calculation~

- **Customer Acquisition Cost / Retention Cost :**

- Marketing spending per customer by events.

- **Get Net Profit Margin to find Net Profit :**

- Among the chosen groups, how much value did they generate (for sales team, we measured the outcome by revenue, for marketing team, profit is a better fit)
- Profit Margin Ratio = Net Income / Net Sales
 - (recall EBITDA from your finance class?)
- Get “Profit Margin Ratio” from the Finance team, if unable, general rules are:
 - 10% for public companies
 - 15% for private companies that are scaling predictably (say above \$10m in ARR, and growing greater than 40% year on year)
 - 20-25% for private companies that have not yet reached scale and predictable growth

CLV – Variables - II

Four major variables continue...

- **Retention Rate / Churn Rate :** (to get Customer Lifespan)
 - Retention Rate = $(CE - CN) / CS \times 100\%$
 - CE: # of customers at the end of a period
 - CN: # of new customers acquired during the period
 - CS: # of customers at the start of a period
 - Churn Rate = $1 - \text{Retention rate}$
 - Customer Lifespan
 - Complete: Use Retention Rate (e.g.// assuming steadily 0.80 (but it can be changed accordingly every year), it will be $0.801 \times 0.802 \times 0.803 \times \dots \times 0.75N$, where N = # of terms, usually monthly, quarterly or yearly)
 - Simple: Use $1/\text{Churn Rate}$, $(1 / (1 - 0.80)) = 5 \text{ years} = \text{Average Customer Lifetime}$
- **Discount Rate :** (Complete Template only)
 - Usually acquired from the Finance department
 - Put inflation rate, currency conversion rate... etc. into account

Templates - Simple

Simple Version (by Year)	
Acquisition Cost Per Customer	\$ 20.00
Total Revenue	\$ 5,000.00
Net Profit Margin	30%
Net Profit	\$ 1,500.00
Total Customers	50
Average Net Profit Per Customer	\$ 30.00
Customers Left	10
Customer Retention Rate	80%
Customer Churn Rate	20%
Average Lifetime in Years	5
Simple CLV	\$ 130.00

- Easy to calculate by inserting necessary data into the green cells
- Less accurate due to assuming fixed churn rate
- Rough estimation

Templates - Complete

Questions. Fill in the yellow boxes and the spreadsheet will take care of the rest.					
	Year 1	Year 2	Year 3	Year 4	Year 5
Customer Segment. How many of a specific group of customers will you start with?	50				
Acquisition Cost? How much did you pay for each new customer? We won't use this figure - see note to explain	\$ 20.00				
Retention Rate. What % of customers will you keep from one year to the next?	80%	80%	80%	80%	80%
Average value. How much is the average revenue, and will this rise over time?	\$ 100.00	\$ 100.00	\$ 100.00	\$ 100.00	\$ 100.00
Net Profit Margin. What % of each order is left after all costs have been accounted for?	30%	30%	30%	30%	30%
Discount Rate. This recognises our money <i>could</i> be better spent on something else - see note to explain	1	1	1	1	1
Answers. These cells will be magically calculated based on the values you put in the table above.					
	Year 1	Year 2	Year 3	Year 4	Year 5
Total Customers. The number of customers at the start of each year from the original segment	50	40	32	26	20
Total Revenue per Customer. This is the total revenue per year for individual customers	\$ 100.00	\$ 100.00	\$ 100.00	\$ 100.00	\$ 100.00
Total Revenue. Annual revenue generated by all the customers in that year	\$ 5,000.00	\$ 4,000.00	\$ 3,200.00	\$ 2,560.00	\$ 2,048.00
Cumulative Revenue. The revenue generated from the (remaining) original customers every year	\$ 5,000.00	\$ 9,000.00	\$ 12,200.00	\$ 14,760.00	\$ 16,808.00
Annual Net profit per customer. Simply, the profit each customer generates in that year.	\$ 10.00	\$ 30.00	\$ 30.00	\$ 30.00	\$ 30.00
Total Net Profit. Profit generated by all the original customers in that year.	\$ 500.00	\$ 1,200.00	\$ 960.00	\$ 768.00	\$ 614.40
Profit at Net Present Value. The profit made each year, even if we offset a better way of spending it!	\$ 500.00	\$ 1,200.00	\$ 960.00	\$ 768.00	\$ 614.40
Cumulative Net Profit at NPV. The profit generated in successive years from the original customers.	\$ 500.00	\$ 1,700.00	\$ 2,660.00	\$ 3,428.00	\$ 4,042.40
Individual LTV at NPV. The cumulative amount of net profit each original customer is worth each year.	\$ 10.00	\$ 34.00	\$ 53.20	\$ 68.56	\$ 80.85

Templates - Complete

- Require more detailed data year by year
- Inserting necessary data into the yellow cells
 - Fixed in the example, but can be adjusted in each year
- Customer values are in an increasing trend, which are closer to real world phenomenon

Conclusion

- CLV will help us answer 3 fundamental questions:
 - Did you pay enough to acquire customers from each marketing channel?
 - Did you acquire the best kind of customers?
 - How much could you spend on retaining customers?
- Too many external factors that possibly change CLV values, therefore, use the tool to help your decision but not depend on it.

The End



Appendix – A

- Worth Reading References:

- <http://www.forentrepreneurs.com/ltv/>

- <https://university.custora.com/for-marketers/clv/basic>

- <https://www.kaushik.net/avinash/analytics-tip-calculate-ltv-customer-lifetime-value/>

- <https://www.datascience.com/blog/intro-to-predictive-modeling-for-customer-lifetime-value>

Appendix – B

- Worth Reading References Continue:

- <http://abovethecrowd.com/2012/09/04/the-dangerous-seduction-of-the-lifetime-value-ltv-formula/>

- https://en.wikipedia.org/wiki/Customer_lifetime_value

- <http://www.forentrepreneurs.com/ltv/>